Legend
⚡ trip: runs fully while the supply received is at least the trip point, else stops
◐ throttle: runs at the share of the supply it receives
▁ floor: on its generator, steps down to the floor share
◷ fallback: full for T hours after telecom is lost, then d_eff
▮ stock: covers its own supply for the hours shown, then runs out

κ: the share of the need the supplier covers. (hidden): a real coupling no source shows; the engine plays it.

FUE-0001
import terminal
not an end use
Needs
◐ power:
POW-0002 κ 1 linear (hidden)
◐ telecom:
TEL-0003 threshold (hidden)
TEL-0001 threshold (hidden)
(2 feeds add, max 1)
◐ transport:
TRN-0001 κ 1 threshold
Back-up
▮ lng: 316.305 h
covers transport; no refill
◷ telecom: full 12 h, then 0.9
Supplies
FUE-0002, POW-0001

FUE-0002
tank farm
not an end use
Needs
⚡ fuel (own network): best of
FUE-0001
⚡ power:
POW-0003 κ 1 linear (hidden)
POW-0002 κ 1 linear (hidden)
(2 feeds add, max 1)
Back-up
▮ oil government: 1513.54 h
covers fuel; no refill
▮ oil industry: 2190.22 h
covers fuel; no refill
Supplies
FUE-0003, FUE-0004

FUE-0003
depot
end use, weight 4
Needs
⚡ fuel (own network): best of
FUE-0002
⚡ power:
POW-0002 κ 1 linear (hidden)
POW-0003 κ 1 linear (hidden)
◐ telecom:
TEL-0001 linear (hidden)
◐ transport:
TRN-0003 κ 1 threshold
Back-up
◷ telecom: full 12 h, then 0.6
Supplies
none

FUE-0004
depot
end use, weight 4
Needs
⚡ fuel (own network): best of
FUE-0002
⚡ power:
POW-0004 κ 1 linear (hidden)
◐ telecom:
TEL-0004 linear (hidden)
◐ transport:
TRN-0003 κ 1 threshold
Back-up
◷ telecom: full 12 h, then 0.6
Supplies
none

HLT-0001
tertiary
end use, weight 10
Needs
▁ to 0.6 power:
POW-0003 κ 1 linear
⚡ telecom:
TEL-0001 linear (hidden)
⚡ water:
WAT-0004 κ 1 linear
◐ transport:
TRN-0002 κ 1 threshold (hidden)
TRN-0001 κ 1 threshold (hidden)
TRN-0003 κ 1 threshold (hidden)
(3 feeds add, max 1)
Back-up
▮ generator fuel: 83.882 h

HLT-0001
tertiary
end use, weight 10
(cont. 2/2)
covers power; refill by delivery
▮ supplies: 114.221 h
covers transport; no refill
▮ water: 57.153 h
covers water; refill: main generator stops after 836.48 h
◷ telecom: full 24 h, then 0.85
Supplies
none

HLT-0002
regional
end use, weight 7
Needs
▁ to 0.6 power:
POW-0002 κ 1 linear
POW-0004 κ 1 linear
(2 feeds add, max 1)
◐ telecom:
TEL-0004 linear (hidden)
TEL-0003 linear (hidden)
⚡ water:
WAT-0004 κ 1 linear
WAT-0003 κ 1 linear
(2 feeds add, max 1)
◐ transport:
TRN-0003 κ 1 threshold (hidden)
TRN-0001 κ 1 threshold (hidden)

HLT-0002
regional
end use, weight 7
(cont. 2/2)
TRN-0002 κ 1 threshold (hidden)
(3 feeds add, max 1)
Back-up
▮ generator fuel: 54.561 h
covers power; refill by delivery
▮ supplies: 33.119 h
covers transport; no refill
▮ water: 107.87 h
covers water; refill: main generator stops after 246.116 h
◷ telecom: full 24 h, then 0.85
Supplies
none

HLT-0003
clinic
end use, weight 3
Needs
⚡ power:
POW-0002 κ 1 linear
◐ telecom:
TEL-0001 linear (hidden)
TEL-0003 linear (hidden)
(2 feeds add, max 1)
⚡ water:
WAT-0003 κ 1 linear
⚡ transport:
TRN-0003 κ 1 threshold (hidden)
Back-up
◷ telecom: full 24 h, then 0.8
Supplies
none

HLT-0004
blood bank
end use, weight 6
Needs
⚡ power:
POW-0003 κ 1 linear
◐ telecom:
TEL-0001 linear (hidden)
◐ transport:
TRN-0002 κ 1 threshold (hidden)
TRN-0003 κ 1 threshold (hidden)
(2 feeds add, max 1)
Back-up
▮ blood: 127.273 h
covers transport; no refill
▮ generator fuel: 37.073 h
covers power; refill by delivery

HLT-0004
blood bank
end use, weight 6
(cont. 2/2)
generator stops after 259.942 h
◷ telecom: full 24 h, then 0.7
Supplies
none

POW-0001
generation
not an end use
Needs
◐ telecom:
TEL-0001 threshold
⚡ water:
WAT-0003 κ 1 linear (hidden)
⚡ fuel:
FUE-0001 κ 1 linear
Back-up
▮ purified water: 10.198 h
covers water; refill: main
◷ telecom: full 12 h, then 0.99
Supplies
POW-0002, POW-0003, POW-0004

POW-0002
transmission substation
not an end use
Needs
◐ power (own network): best of POW-0001
◐ telecom:
TEL-0001 threshold
Back-up
◷ telecom: full 12 h, then 1
manual switching if telecom is down: 1 h on site
Supplies
HLT-0002, HLT-0003, POW-0003 POW-0004, TEL-0002, TEL-0003 TEL-0004, WAT-0002, WAT-0003

POW-0003
distribution substation
end use, weight 5
Needs
◐ power (own network): best of POW-0001, POW-0002
◐ telecom:
TEL-0003 threshold
TEL-0004 threshold
(2 feeds add, max 1)
Back-up
◷ telecom: full 12 h, then 1
manual switching if telecom is down: 1 h on site
Supplies
HLT-0001, HLT-0004, TEL-0001 WAT-0003

POW-0004
distribution substation
end use, weight 5
Needs
◐ power (own network): best of POW-0001, POW-0002
⚡ telecom:
TEL-0001 threshold
TEL-0003 threshold
Back-up
◷ telecom: full 12 h, then 1
manual switching if telecom is down: 1 h on site
Supplies
HLT-0002, TEL-0002, TEL-0004

TEL-0001
exchange
not an end use
Needs
⚡ power:
POW-0003 κ 1 linear
Back-up
▮ battery: 2.393 h
covers power; refill: battery
▮ generator fuel: 26.38 h
covers power; refill by delivery
generator stops after 2094.77 h
Supplies
POW-0001, POW-0002, POW-0004 TEL-0002, TEL-0003, TEL-0004

TEL-0002
mast
end use, weight 1
Needs
⚡ telecom (own network): best of
TEL-0001
⚡ power:
POW-0002 κ 1 linear
POW-0004 κ 1 linear
Back-up
▮ battery: 3.901 h
covers power; refill: battery
Supplies
none

TEL-0003
mast
end use, weight 1
Needs
⚡ telecom (own network): best of
TEL-0001
⚡ power:
POW-0002 κ 1 linear
Back-up
▮ battery: 7.725 h
covers power; refill: battery
Supplies
POW-0003, POW-0004

TEL-0004
mast
end use, weight 1
Needs
⚡ telecom (own network): best of
TEL-0001
⚡ power:
POW-0004 κ 1 linear
POW-0002 κ 1 linear
(2 feeds add, max 1)
Back-up
▮ battery: 4.358 h
covers power; refill: battery
Supplies
POW-0003

TRN-0001
seaport
end use, weight 7
Needs
⚡ power:
POW-0002 κ 1 linear
(hidden)
◐ telecom:
TEL-0001 linear (hidden)
◐ fuel:
FUE-0003 κ 1 threshold
(hidden)
FUE-0002 κ 1 threshold
(hidden)
(2 feeds add, max 1)
Back-up
◷ telecom: full 24 h, then 0.5
Supplies
FUE-0001

TRN-0002
airport
end use, weight 6
Needs
⚡ power:
POW-0002 κ 1 linear
(hidden)
POW-0003 κ 1 linear
(hidden)
(2 feeds add, max 1)
⚡ telecom:
TEL-0001 threshold (hidden)
⚡ fuel:
FUE-0002 κ 1 linear
(hidden)
Back-up
◷ telecom: full 0 h, then 0.2
Supplies
none

TRN-0003
freight yard
end use, weight 4
Needs
⚡ telecom:
TEL-0002 linear (hidden)
TEL-0001 linear (hidden)
(2 feeds add, max 1)
◐ fuel:
FUE-0002 κ 1 threshold
(hidden)
Back-up
◷ telecom: full 24 h, then 0.5
Supplies
FUE-0003, FUE-0004

TRN-0004
bridge
not an end use
Needs
nothing
Back-up
none
Supplies
none

WAT-0001
reservoir
not an end use
Needs
nothing
Back-up
none
Supplies
WAT-0002

WAT-0002
treatment
not an end use
Needs
◐ water (own network): best of
WAT-0001
◐ power:
POW-0002 κ 1 linear
◐ telecom:
TEL-0001 threshold (hidden)
TEL-0004 threshold (hidden)
(2 feeds add, max 1)
Back-up
▮ generator fuel: 35.394 h
covers power; refill by delivery
generator stops after 3535.32 h
◷ telecom: full 24 h, then 0.95

WAT-0002
treatment
not an end use
(cont. 2/2)
Supplies
WAT-0003

WAT-0003
pumping
not an end use
Needs
◐ water (own network): best of
WAT-0002
◐ power:
POW-0003 κ 1 linear
POW-0002 κ 1 linear
(2 feeds add, max 1)
◐ telecom:
TEL-0001 threshold (hidden)
Back-up
◷ telecom: full 12 h, then 0.9
Supplies
HLT-0002, HLT-0003, WAT-0004

WAT-0004
storage
end use, weight 5
Needs
⚡ water (own network): best of
WAT-0003
Back-up
▮ distribution water: 24.011 h
covers water; refill: pumps
Supplies
HLT-0001, HLT-0002